

Towards Emergency Scenarios: An Integrated Decision-Making Framework of Multi-Lane Platoon Reorganization

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Abstract—Vehicle platoons are considered an effective strategy to enhance road safety and traffic efficiency, yet their capability to respond to sudden emergencies remains insufficiently addressed. Existing studies mainly focus on stable formation control or coordination in normal conditions, while systematic frameworks for dynamic platoon reorganization in high-risk scenarios are lacking. To fill this gap, this article proposes a platoon distribution reorganization decision-making framework that integrates three layers: platoon distribution, vehicle cooperative decision-making, and vehicle planning and control. First, a reinforcement-learning-based distribution model is developed, in which a risk potential field quantitatively evaluates driving risks and a tailored reward function guides the reorganization process. Second, a coalition-game-based decision-making model captures inter-vehicle cooperation by dynamically forming coalitions and generating optimal actions. Moreover, a novel graph-theory-based Platoon Disposition Index (PDI) is introduced into the reward function to evaluate platoon distribution states and accelerate reorganization. Finally, validation under two high-risk scenarios with random traffic flows demonstrates that the proposed framework significantly reduces collision risk and improves traffic efficiency, while the inclusion of PDI further shortens reorganization time and enhances overall efficiency.

Index Terms—Vehicle platoon, Autonomous driving, Decision and control, Platoon reorganization.

I. INTRODUCTION

PLATOONS of Intelligent and connected vehicles can potentially improve traffic efficiency by organizing vehicles in small spaces and improving fuel economy by reducing air resistance [1], [2]. However, existing platoon control strategies still encounter significant safety risks in practical applications, particularly due to their insufficient capability in addressing sudden hazardous situations. For instance, when unexpected obstacles emerge ahead or when the platoon

formation is disrupted by cut-in vehicles, the coordination among vehicles is often not achieved in a timely manner. Consequently, safe obstacle avoidance within limited road space cannot be guaranteed, which in turn diminishes the overall safety of platoon driving [3].

The conventional platoon control literature typically assumes fixed single-file formations and emphasizes rapid and precise regulation of inter-vehicle spacing. Approaches that adopt a leader follower architecture or a virtual structure architecture, as reported in [4], rely on a designated leader to issue commands while follower vehicles estimate relative deviations and apply tracking controllers to reduce those deviations. However, fixed formations are ill suited to mixed traffic because they require larger longitudinal gaps and longer maneuvering time, they impede timely avoidance of dynamic obstacles such as a fast approaching vehicle from behind, and they increase the likelihood of rear-end collisions owing to delayed lane change responses. Therefore, adopting dynamic platoon configurations and enabling real-time reconfiguration according to prevailing traffic conditions is necessary to enhance safety and travel efficiency.

The rule-based method employs finite-state logic to switch among a limited set of fixed formations, such as single-lane, two-lane and three-lane configurations. This approach represents a straightforward implementation of adaptive platoon reconfiguration. However, fixed schemes are generally unsuitable for highly dynamic, time-varying traffic. Their limited mode set and rigid switching logic cannot respond adequately to continuously evolving conditions. Consequently, formation-control methods that support real-time adaptation to changing traffic states are required.

The optimization-based methods facilitate the computation of optimal, safe, feasible and collision-free trajectories for each vehicle in the platoon. Modeling vehicle trajectories through constrained centralized optimization permits flexible lane changes and formation reconfiguration, thereby enhancing safety, traffic-flow efficiency and ride comfort. However, formulating such multi-agent optimization problems is particularly challenging in mixed traffic. Vehicle heterogeneity, partial observability and complex interaction dynamics hinder accurate centralized modeling. Furthermore, highly detailed models impose substantial computational burdens that scale rapidly with platoon size and undermine the timeliness of real-time decision making. This limitation is especially critical in emergency scenarios involving high-speed dynamic obstacles [5].

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Multi-agent reinforcement learning (MARL) holds considerable promise for addressing complex traffic scenarios and improving real-time responsiveness [6], [7]. In the context of platoon decision-making and control, it offers the capability to support decentralized coordination, scalable learning, and adaptive responses to dynamic environments. However, the predominant reliance on neural networks as non-transparent function approximators reduces interpretability and impedes the provision of formal safety guarantees for vehicle operation. Moreover, limited attention has been given in contemporary MARL frameworks to inter-agent social considerations. Mechanisms for equitable allocation of benefits among agents are often absent, and the reconciliation of individual vehicle objectives with collective platoon goals remains insufficiently addressed.

In this article, a Game/RL dual-layer decision-making and motion planning framework (GRDF) is proposed for multilane autonomous vehicle platoons to enable reorganization in emergency environments. The framework integrates learning-based and game-theoretic methods and simplifies the reorganization problem under time-varying traffic through a dual-layer optimization model. The deep reinforcement learning module captures the dynamic characteristics of traffic flow and generates high-level task allocations, while the game-theoretic strategy balances individual and collective interests during task assignment, mitigating unequal benefit distribution among agents and enhancing the sociality of decision execution. This two-layer design improves overall safety and fault tolerance, with strategic task assignment handled by the learning layer and safety-critical collision-avoidance maneuvers computed by the game-theoretic layer.

The main contributions of this article are summarized as follows:

- i) An upper-layer RL-based model is developed to determine appropriate platoon formations according to environmental risk levels. A risk potential field is incorporated into the reward function to explicitly evaluate exploration risks and enhance model safety.
- ii) A lower-layer coalition-game-based model is proposed to generate optimal behaviors for each vehicle within the current formation. A graph-theory-based index Platoon Disposition Index (PDI) is introduced to evaluate platoon distribution, which has been shown to reduce reorganization time, improve efficiency, and enhance overall traffic performance.
- iii) A dual-layer RL-game decision-making and motion planning framework is presented for platoon reorganization. Its effectiveness is demonstrated in multiple highway emergency scenarios with uncertainty, showing improvements in both safety and traffic efficiency.

The remainder of this article is as follows: Section II introduces related works. Section III establishes the overall framework of the platoon reorganization strategy. The method of making platoon-layer decisions and vehicle-layer decisions is represented in Section IV and V. Section VI designs experiments involving multiple scenarios and thoroughly validates the effectiveness of the proposed strategy. Section VII draws a clear conclusion of this article.

II. RELATED WORKS

The related works are summarized from two aspects, the P-CLCC strategies for multi-lane vehicle platoons and the cooperative decision-making methods of intelligent vehicle platoons.

A. P-CLCC Strategies

The schemes of the P-CLCC methods can be divided into two main types.

i) Decoupling decision making and control for the vehicle platoon: This scheme usually adopts a rule-based model to make the lane change decisions of each vehicle in a certain order [8]. When the lane change space for the vehicle platoon is large enough, the vehicles are able to make lane change decisions together [9], [10]. When there is not enough free space, work [11] proposed a successive platoon lane change approach to improve the passing ability of the platoon in a high-density traffic environment. Moreover, works [12], [13], [14] put forward a particular P-CLCC logic that, the vehicle in platoon, who satisfies the safety conditions at the first place, should change to the target lane immediately, and then slow down to make enough space for other vehicles within a given time window. However, the predefined rules or logic for lane changing are not suitable for complex environments, especially when encountering aggressive human drivers or sudden emergency situations.

ii) Coupling decision making and control for the vehicle platoon: This scheme allows vehicles in the platoon to make lane change decisions asynchronously, which can coordinate all platoon vehicles to a greater degree when faced with more complex and emergency environments. Most studies transform the cooperative lane change process into a Multiple Vehicle Motion Planning (MVMP) problem [15], [16], [17]. Work [16] proposed a MVMP method for platoon using Monte Carlo Tree Search, which splits the whole process into a series of lane change tasks and determines the optimal lane change sequence. The works [18], [19], [20] established a central Mixed Integer Linear Programming (MILP) model to calculate each vehicle's optimal discrete lane change decision. However, the MVMP model mainly focus on the trajectory planning process and it still needs clear inputs of starting and ending states of vehicles. Studies such as [21], [22], [23], [24] proposed multilayer decision structures, containing the upper platoon coordination layer and the lower vehicle motion planning layer, which can leverage the benefits of vehicle collaboration and improve model generalization. Works [21] and [22] presented a optimal-based model to generate the ideal formation order and motion planning was executed via Model Predictive Control (MPC) system. The dual-layer decision-making framework significantly outperforms the MVMP framework in terms of generalization and computational efficiency. However, pure optimization-based frameworks require the formulation of complex multi-variable and multi-objective coupled optimization problems, facing both solution difficulty and computational efficiency challenges.

B. Cooperative Decision-Making Methods of Multi-Lane Platoons

Existing Cooperative decision-making methods for autonomous vehicle platoons can generally be categorized into learning-based methods, optimization-based methods, and other approaches. Among the optimization-based methods, there are further distinctions between pure optimization and game-theoretic optimization approaches.

i) Learning-based methods: Longitudinal control of vehicle platoons using reinforcement learning or multi-agent reinforcement learning has become increasingly prevalent [25], [26], [27], [28], [29]. Work [26] proposes a longitudinal control algorithm for platooning based on multi-agent reinforcement learning, which allocates reward functions according to the car-following relationships within the platoon to address the “lazy agent” problem. Work [28] introduces a multi-agent reinforcement learning algorithm aimed at enhancing safety in human-machine mixed platoons. The approach incorporates a CEF-QP constraint and an uncertainty-aware trajectory prediction model to improve control safety. However, these researches only account for uncertainties within the platoon, such as driver reaction times, while neglecting external uncertainties, such as obstacles ahead or sudden intrusions from other vehicles. Moreover, these learning-based methods have poor ability to efficiently coordinate the lateral maneuvers of the platoon, lack of adaptability to complex scenarios.

ii) Optimization-based methods: Optimization-based and game theory-based methods are primarily applied to longitudinal and lateral cooperative control of vehicle platoons, collaborative obstacle avoidance decision-making, and decisions related to vehicle merging into or leaving the platoon [30], [31], [32]. Work [33] proposes a non-cooperative game decision-making model for platoons to handle scenarios involving lane-right conflicts. When an external vehicle attempts to cut into the platoon, the model determines the entry position of the vehicle within the platoon through a non-cooperative game, automatically adjusting the longitudinal gaps in the platoon. This method effectively mitigates collision risks caused by external vehicle cut-ins. However, the platoon passively accommodates the cut-in behavior, leading to disruption of its formation. Over time, this could result in reduced driving efficiency for the platoon. Work [34] introduces a game-theory-based cooperative obstacle avoidance method for vehicle platoons. This approach assumes that the platoon can evade obstacles and execute overtaking maneuvers by transitioning between predefined configurations. However, the method only considers a limited number of fixed configurations, and the transitions between these configurations are governed by simple logical rules. Consequently, the approach has limited adaptability to diverse and complex scenarios, as the transition rules lack sophistication and generalization capabilities.

iii) Other methods: Work [35] proposes a rule-based and graph-theoretic approach for cooperative obstacle avoidance and decision-making in platoons. When obstacles are detected, the leading vehicle guides the following vehicles to perform

formation adjustments by navigating through graph nodes to avoid the obstacle. Work [36], [37] introduces a multi-mode platoon decision control algorithm, dividing platoon control into functions such as cruising, following, lane-changing, overtaking, and parking. Mode transitions are managed via a state machine, aiming to enhance the adaptability of the platoon to diverse scenarios. However, most methods rely solely on physical rules tailored to specific scenarios, limiting their scalability and adaptability to dynamic environments. What's more, when external risks disrupt the platoon formation, these methods fail to provide effective countermeasures to restore the formation or mitigate the resulting safety risks.

III. OVERALL FRAMEWORK

A. Problem Statement

Assume the traffic flow condition is a mixed traffic flow: the vehicle platoon consists of Connected and Autonomous Vehicles (CAVs), while the background traffic flow consists of Human-Driven Vehicles (HDVs). Within the platoon, vehicles share state information, which is transmitted following a specified communication topology. The traffic flow set is noted as V .

$$V = \{V_1^c, V_2^c, V_3^c, V_4^h, V_5^h, \dots\} \quad (1)$$

The platoon adopts a Leader-Leader-Predecessor-Follower (LLPF) communication topology, defined as follows: There is bidirectional communication between different platoon leaders. There is bidirectional communication between the leader and its directly connected followers within the platoon and between each follower and its immediate predecessor. The schematic diagram of the communication topology adopted by the platoon is shown in Figure 1. The information exchanged between vehicles is represented in (2).

$$F = \{l_{target,i}, x_i, y_i, v_{x,i}, v_{y,i}, a_{x,i}, a_{y,i}\} \quad (2)$$

where $l_{target,i}$ means the target lane number of the i -th vehicle, $x_i, y_i, v_{x,i}, v_{y,i}, a_{x,i}$ and $a_{y,i}$ denote the position, speed and acceleration information of the i -th vehicle.

The communication topology of the vehicle platoon is modeled on the basis of graph theory, noted as $G = (V^c, E)$. E is the set of edges between vehicles.

Definition 1 (Platoon Communication Topology): For $\forall i, \in [0, n-1], \forall j, \in [1, n], E_{ij}$ satisfies:

$$E_{ij} = \begin{cases} 1 & \text{if } i = 0 \text{ or } i < j \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where n represents the number of platoon vehicles, $0 \sim n$ means the numbering of all vehicles in the platoon in descending order based on their longitudinal positions. $E_{ij} = 1$ denotes the information can be transferred from V_i^c to V_j^c .

To reduce additional computational complexity, the vehicles within the platoon are simplified using a kinematic model.

$$\begin{aligned} \dot{y}(t) &= u(t) \cos(\theta(t)) \\ \dot{x}(t) &= u(t) \sin(\theta(t)) \end{aligned} \quad (4)$$

where $u(t)$ means vehicle speed, $\theta(t)$ indicates vehicle driving heading angle.

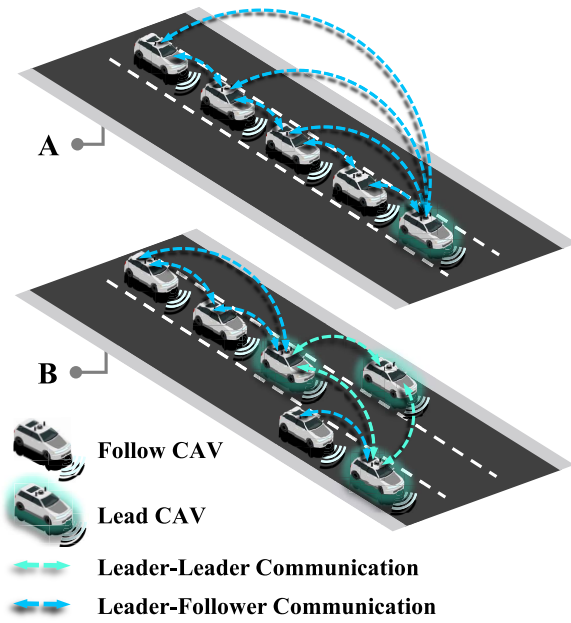


Fig. 1. The communication topology of vehicle platoon.

B. The Dual-Layer Framework

In this article, we present a Game/RL Dual-layer driven decision making and motion planning Framework (GRDF) for multilane autonomous vehicle platoons in highway environments. The diagram of GRDF is shown in Figure 2.

The GRDF framework proposed in this article is divided into three main modules: the platoon distribution model, the vehicle cooperative decision-making model, and the vehicle execution model.

Environmental information including around vehicle states and risk, is input to the platoon distribution layer, and the output is the ideal platoon configuration. In this model, an RL-based decision-making method is proposed, which extracts necessary observation information from the dynamic environment and calculates the desired platoon grouping using an actor-critic network architecture.

After receiving the platoon distribution behavior, the vehicle decision layer is supposed to generate the behavioral decision-making results for each vehicle in platoon. In this model, we propose a coalition game-based cooperative decision-making method and formulate an mixed-integer optimization problem (MIP) optimization problem to compute the game equilibrium solution. Additionally, to improve optimization efficiency, a pruning strategy is designed to eliminate actions that are infeasible or do not meet safety requirements.

Finally, the vehicle execution model is created to transform the vehicle behavior into the longitudinal and lateral control information and make the vehicle drive in obedience to the given control commands. In this model, the motion planning is realized through lattice method. The trajectory clusters are generated using polynomials and a dynamics stability checker id designed to validate the trajectories. The best trajectory is selected by a comprehensive evaluation function, considering efficiency, safety, comfort and so on. The lower-level vehicle following and steering control are achieved using linear

quadratic regulator (LQR) and Proportional Integral Derivative (PID) controllers.

IV. PLATOON DISTRIBUTION LAYER

A. Preliminary Knowledge

Reinforcement learning methods are based on Markov processes, where reward or penalty functions are predefined. Through training, agents interact with the environment to learn optimal strategies that maximize the reward or minimize the penalty function. Deep reinforcement learning further facilitates approximating highly nonlinear functions from complex datasets, making it well-suited for addressing complex decision-making and control problems [38].

Early reinforcement learning algorithms, such as Deep Q Network (DQN), use neural networks to predict the Q-values of actions, aiming to minimize the discrepancy between the predicted Q-value network and the target Q-value network. The loss function can be defined as (5). However, the process of optimizing and updating the action network may not align with the dynamic characteristics of the environment, presenting challenges for applications in complex and dynamic scenarios.

$$L(\theta) = \mathbb{E}_{(s,a,r,s')} \left[(r + \gamma)^2 \right] \max_{a'} Q_{\theta_{target}}(s', a') \quad (5)$$

where (s, a, r, s') represents the transition tuple, s is the current state, a is the action taken, r is the reward received, and s' is the next state, $Q_{\theta}(s', a')$ is the Q-value estimated by the current network with parameters θ , $Q_{\theta_{target}}(s', a')$ means the target Q-value estimated by the target network with parameters θ_{target} , γ denotes the discount factor, which determines the weight of future rewards.

The Proximal Policy Optimization (PPO) algorithm is implemented based on the actor-critic architecture. It approximates the optimal policy using the actor network and evaluates the policy's Q-values through the critic network, integrating policy gradient and value network update methods for more accurate policy evaluation and updates. Additionally, the PPO method introduces a clipping function to constrain the range of policy update gradients, thereby enhancing the stability of the learning process. The loss function for policy gradient updates is defined as (6).

$$L^{FG}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (6)$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio between the new policy π_{θ} and the old policy $\pi_{\theta_{old}}$, $\hat{A}_t$ represents the advantage estimate at time step t , and ϵ is a hyperparameter controlling the clipping range to balance exploration and stability during policy updates.

B. Centralized Proximal Policy Optimization Algorithm

In this section, we propose a platoon configuration decision-making method based on the PPO algorithm. The leading vehicle can obtain the state and perception information of other vehicles through the communication topology, enabling information sharing within the platoon. Based on this

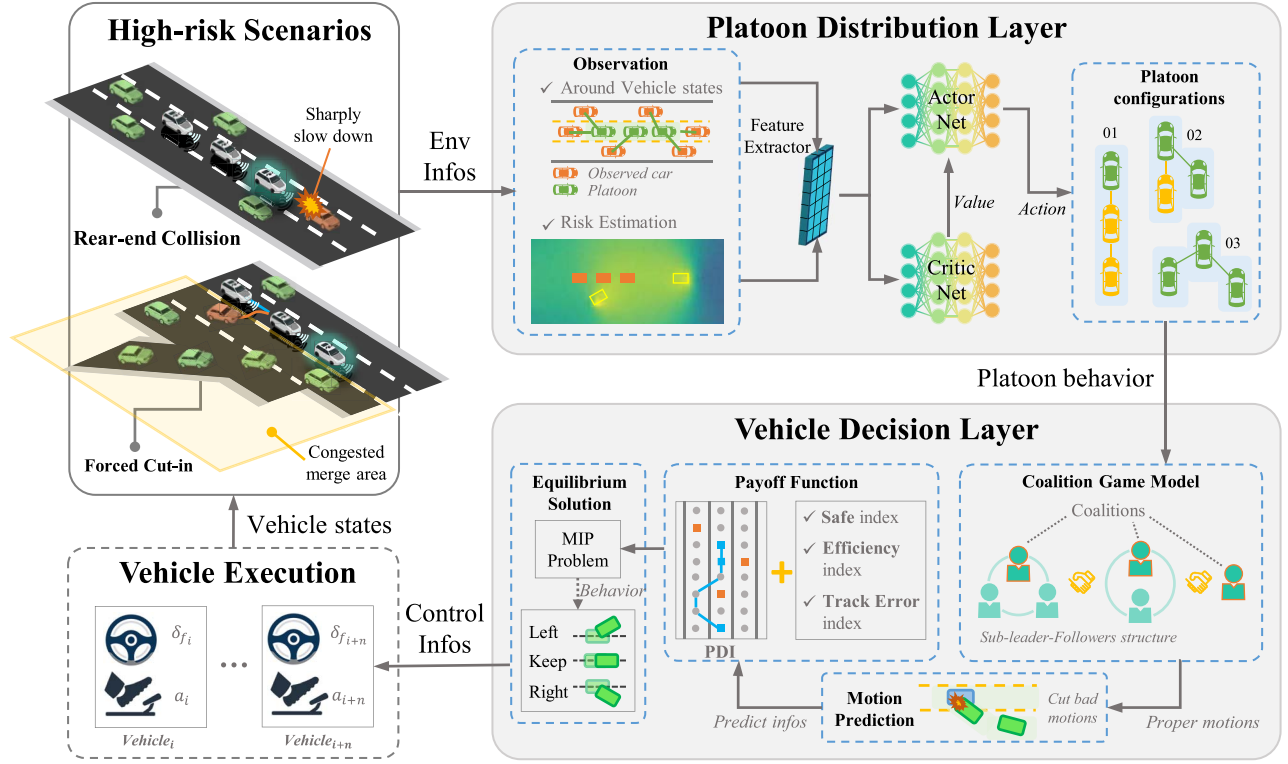


Fig. 2. The overall structure of GRDF framework.

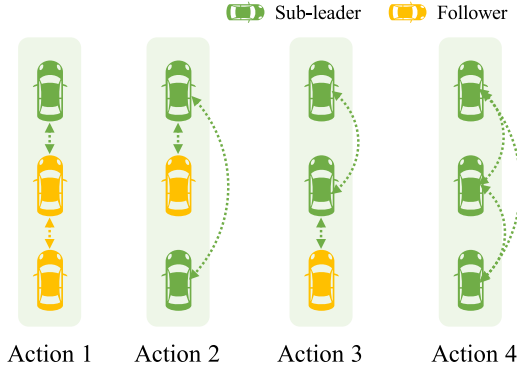


Fig. 3. The platoon configuration actions.

information-sharing mechanism, the leading vehicle performs centralized decision-making for platoon configuration.

The decision-making process is modeled as a general Partial Observation Markov Decision Process (POMDP), $(S, A, P, R, \gamma, O, Z)$. Among these elements, the action space A , observation space O , and reward function R are particularly important.

The action space A includes various platoon configurations, defined in (7). Taking 3-vehicle platoon as an example, the platoon configuration action is illustrated in Figure 3.

$$\begin{aligned} A &= \{a_1, a_2, a_3, \dots, a_m\} \\ a_1 &= [(0, 1, 2, \dots, m)] \\ a_2 &= [(0)(1, 2, \dots, m)] \\ a_3 &= [(0)(1)(2, 3, \dots, m)] \end{aligned} \quad (7)$$

where a_i represents each platoon configuration. m is the total number of a n -vehicle platoon's configuration, $m = C_{n-1}^1 + C_{n-1}^2 + \dots + C_{n-1}^{n-1} + 1$.

The observation space O denotes the observed environment information accessible to the controlled platoon, including positions, speeds, accelerations and jerk of objects. Additionally, we incorporated the TTC index into the observation space as an explicit representation of environmental risk states. This enhancement improves the algorithm's adaptability to high-risk scenarios by providing a quantitative measure of potential collision risks. The observations are obtained through sensors and communication units with random noise. The observation space O is defined in (8).

$$O = \{x, y, v_x, v_y, a_x, a_y, j_x, j_y, \tau\} \quad (8)$$

where $x, y, v_x, v_y, a_x, a_y, j_x, j_y$ represent longitudinal and lateral position, speed, acceleration and jerk of observed objects. The values of a_x, a_y, j_x, j_y are derived by v_x, v_y . τ means the TTC between ego vehicle and observed objects.

The reward function R guides the training process by defining the awards and penalties for decision-making results. The reward function primarily considers safety R_s , efficiency R_e , platoon driving performance R_d and platoon reorganization performance R_r .

$$R = w_s \times R_s + w_e \times R_e + w_d \times R_d + w_r \times R_r \quad (9)$$

The safety reward R_s consists of the collision penalty r_{col} and the risk penalty r_{ris} .

$$R_s = w_{col} \times r_{col} + w_{ris} \times r_{ris} \quad (10)$$

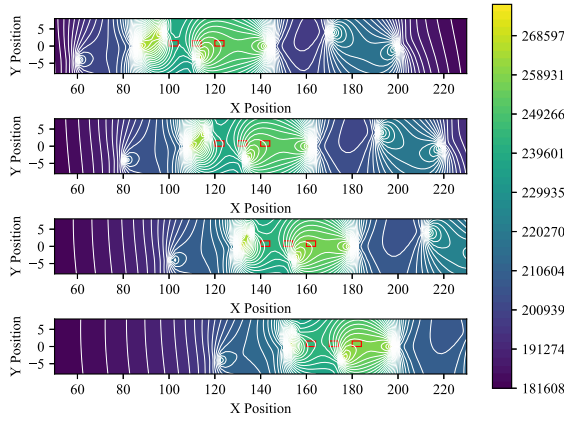


Fig. 4. The risk field distribution.

$$r_{col} = \begin{cases} 0, & \text{if collision happened} \\ 1, & \text{else} \end{cases} \quad (11)$$

The risk reward r_{ris} is characterized by the field intensity of the driving risk field, determined in (12). The detailed construction of the risk field refers to work [39], the scenario risk distribution is shown in Figure 4.

$$r_{ris} = \max_{i \in [1, n]} \left(\frac{GRM}{d_i^{k_1}} \right)^{k_2 v_i} / \left(\frac{GRM}{d_{\min}^{k_1}} \right)^{k_2 v_{\max}} \quad (12)$$

The efficiency reward R_e is the platoon's average speed.

$$R_e = \frac{1}{n} \sum_{i=1}^n \frac{v_i}{v_{\max}} \quad (13)$$

The platoon driving performance reward R_d contains the tracking error of distance, lateral position and speed.

$$R_d = \frac{1}{n-1} \sum_{i=1}^{n-1} w_x \times (x_i - x_{i+1} - d_{\text{target}}) + w_y \times (y_i - y_{i+1}) + w_v \times (v_i - v_{i+1}) \quad (14)$$

where the target follow distance d_{target} in platoon is 10 m.

The platoon reorganization performance reward R_r is composed of platoon reorganization frequency r_{rf} , reorganization time r_{re} and reorganization incentive r_{re} , shown as follows

$$R_r = w_{rf} \times r_{rf} + w_{re} \times r_{re} + w_{ri} \times r_{ri} \quad (15)$$

The platoon reorganization frequency reward r_{rf} means the number of platoon reorganization events triggered per unit time, which is defined and normalized in (16). When a platoon reconfigures, its aggregate aerodynamic drag typically increases, degrading fuel efficiency. Moreover, excessively frequent formation changes may perturb surrounding traffic and even induce localized flow oscillations. Consequently, our control objective—subject to maintaining platoon safety—is to minimize the incidence of reconfiguration events and to reduce the duration of each reconfiguration.

$$r_{rf} = \frac{N_{\text{trigger}}}{N_{\text{step}}} \quad (16)$$

where N_{trigger} refers to the decision steps involved in switching the platoon from a single sub-platoon configuration to other multiple sub-platoon configurations during a single episode. N_{step} represents the total decision steps within a single episode.

The platoon reorganization time r_{re} represents the average time taken by the platoon to successfully complete a reorganization process. Successfully completing a reorganization process means that the platoon switches from a single sub-platoon configuration to a multi-sub-platoon configuration and then back to a single sub-platoon configuration.

The platoon reorganization incentive r_{rc} denotes the degree of incentive for triggering platoon reorganization behavior under varying environments. When the platoon encounters a sudden decrease in environmental risk, denoted by TTC or a significant decline in efficiency, denoted by R_e , higher reorganization incentive is provided. Conversely, when r_{ris} remains stable and the R_e keeps in a higher level, a lower reorganization incentive is applied.

$$r_{rc} = k_t \times \frac{\tau_{\min}^0}{\tau^0} + k_v \times \sum_{i=1}^n \frac{v_i}{v_{\max}} \quad (17)$$

where τ^0 represents the TTC between platoon leader and front object, $\tau_{\min}^0$ is the critical value of TTC, set as 2.5 s. k_t , k_v are constant parameters.

V. VEHICLE DECISION LAYER

After the platoon-layer decision-making model calculates the optimal platoon configuration, this section establishes a coalition game model to determine the behavioral decisions for each vehicle.

A. Formulation of Coalition Game Decision-Making Model

Coalition game is a type of cooperative game, which usually involves multiple players, denoted by $\mathcal{N} = \{1, \dots, n\}$, who seek to form cooperative groups to achieve the optimal collective payoff distribution. Each group is called a coalition. Players within a coalition reach an agreement and act as a unified entity in the collaborative decision-making process. v represents the quantified benefit that the coalition achieves during the game. The definition of v determines the specific form and type of the game. Hence, the coalition game can be described by a pair $(\mathcal{N}, v)$.

The characteristic function v can be defined in two ways [40]. In coalitional games with nontransferable utility (NTU), the payoff of each coalition S is closely related to the joint actions of other coalitions. In games with transferable utility (TU), however, the payoff of each coalition is independent of the performance of other coalitions.

Definition 2 (The Characteristic Function v in NTU Games): In a NTU game $G = (\mathcal{N}, v)$, coalition S is a subset of $\mathcal{N}$. All coalitions other than S are supposed to form a complementary coalition, denoted as $\mathcal{N} \setminus S$. Treat S and $\mathcal{N} \setminus S$ as two players participating in a non-cooperative game. The actions of coalition is denoted by a , and the profit of actions is denoted by u . The characteristic function v of NTU game can be calculated by the maximin principle in (18), which means

that the sum of the maximum payoffs guaranteed to coalition S members when facing the most severe threats from coalition $\mathcal{N} \setminus S$ is ensured.

$$v(S) = \min_{\mathcal{N} \setminus S} \max_S \sum_{i \in S} u_i(a_S, a_{\mathcal{N} \setminus S}) \quad (18)$$

where a_S and $a_{\mathcal{N} \setminus S}$ means the actions of coalition S and complementary coalition $\mathcal{N} \setminus S$. Natural convention $V(\emptyset) = 0$.

Definition 3 (The Characteristic Function v in TU Games): In a TU game $G = (\mathcal{N}, v)$, coalition S is a subset of $\mathcal{N}$. The actions of coalition is denoted by a , and the profit of actions is denoted by u . The characteristic function v of TU game can be defined in (19).

$$v(S) = \max_S \sum_{i \in S} u_i(a_S, a_{\mathcal{N} \setminus S}) \quad (19)$$

The NTU games contains implicit competitive relationship among coalitions. While in TU games, coalitions are in a totally cooperative relationship. In this article, all vehicles within the platoon are committed to cooperation, thus a TU game is established.

In our platoon reorganization problem, the coalition S needs to satisfy follow conditions.

Remark 1 (Vehicle Coalition Formation Conditions): When the vehicles drive on the same lane and no other obstacles exist between them, a coalition can be formed from these vehicles. The positions of these vehicles are limited by (20). Each coalition is a sub-platoon. The vehicle at the forefront becomes the leader of the sub-platoon. Based on the LLPF communication topology, the vehicles within the coalition and the sub-leaders between coalitions can exchange driving information.

$$\begin{aligned} \max_i |x_i - x_{i+1}| &< \tilde{x}_{lim} \\ \max_i |y_i - y_{i+1}| &< \tilde{y}_{lim} \end{aligned} \quad (20)$$

where $\tilde{x}_{lim}$ and $\tilde{y}_{lim}$ are constants, $\tilde{x}_{lim} = 30$ m and $\tilde{y}_{lim} = 1.5$ m.

B. Coalition Characteristic Function

The goal of platoon decision-making is to ensure that each sub-platoon, or coalition, completes the configuration transformation process safely and efficiently. This process can be divided into two phases: platoon splitting and platoon merging.

The split state occurs when the platoon divides into sub-platoons to avoid obstacles, whereas the merge state denotes their rejoining into the original formation. Figure 5(c) illustrates this process: the transition from the left to the right formation involves both split and merge. Within each (sub-)platoon, communication is unidirectional, while leaders maintain bidirectional links. During splitting, the sub-platoon of Vehicles 1–2 separates and its link is disconnected. Vehicle 2 then forms a new sub-platoon with Vehicle 3, establishing a new link. As Vehicle 3 ceases to be a leader, its bidirectional link with Vehicle 1 is also removed.

The characteristic function $v_p(S)$ of platoon splitting and platoon merging process is shown in (21).

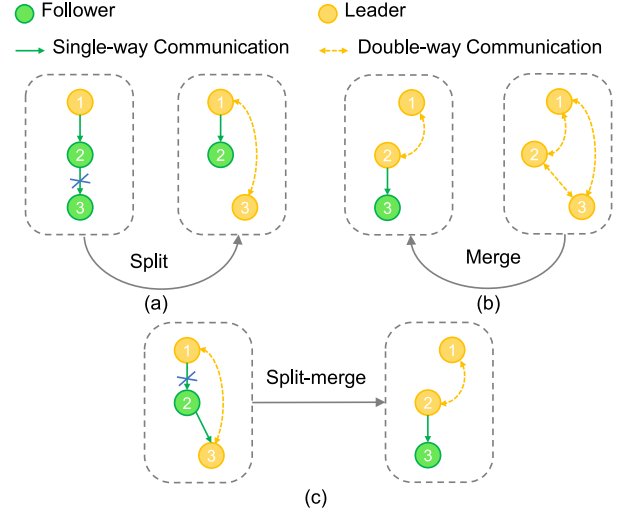


Fig. 5. Schematic of platoon split and merge.

$$v_p(S) = \max_S \sum_{i \in S} J_{sp}^i(\alpha_i, \beta_i) \quad (21)$$

$$v_p(S) = \max_S \sum_{i \in S} J_{mg}^i(\alpha_i, \beta_i)$$

where $J_{sp}(\alpha_i, \beta_i)$ and $J_{mg}(\alpha_i, \beta_i)$ are the profit of platoon splitting and platoon merging process. α_i and β_i are the i -th action in a_S and $a_{\mathcal{N} \setminus S}$, the action list of current coalition S and the complementary coalition $\mathcal{N} \setminus S$. The action α, β refers to the lateral decision of each vehicle, $\alpha, \beta \in \{\text{left lane change, right lane change, keep the lane}\}$. Each vehicle in platoon is equipped with advanced driving assistance functions, including Adaptive Cruise Control (ACC) and Automated Lane Change (ALC): lane changes (“left” or “right”) are executed via the ALC controller, while lane keeping is governed by the ACC controller. In this article, the ACC process is implemented using a Linear Quadratic Regulator (LQR) algorithm, and the ALC process is realized through the Minimizing Overall Braking Induced by Lane Change (MOBIL) method.

The splitting profit $J_{sp}(\alpha_i, \beta_i)$ includes safety profit $J_s(\alpha_i, \beta_i)$, driving efficiency profit $J_e(\alpha_i, \beta_i)$ and platoon integration profit $J_{it}(\alpha_i, \beta_i)$. The merging profit $J_{mg}(\alpha_i, \beta_i)$ contains formation error $J_{tr}(\alpha_i, \beta_i)$ on the basis of the splitting profit $J_{sp}(\alpha_i, \beta_i)$. $J_{sp}(\alpha_i, \beta_i)$ and $J_{mg}(\alpha_i, \beta_i)$ are calculated by (22).

$$\begin{aligned} J_{sp}(\alpha_i, \beta_i) &= w_s \times J_s(\alpha_i, \beta_i) + w_e \times J_e(\alpha_i, \beta_i) \\ &\quad + w_{it} \times J_{it}(\alpha_i, \beta_i) \\ J_{mg}(\alpha_i, \beta_i) &= w_s \times J_s(\alpha_i, \beta_i) + w_e \times J_e(\alpha_i, \beta_i) \\ &\quad + w_{it} \times J_{it}(\alpha_i, \beta_i) \\ &\quad + w_{er} \times J_{er}(\alpha_i, \beta_i) \end{aligned} \quad (22)$$

where w_s, w_e, w_{it} and w_{er} are constant weights.

The safety profit $J_s(\alpha_i, \beta_i)$ is calculated in (23).

$$\begin{aligned} J_s(\alpha_i, \beta_i) &= \left(P_{ris}(\alpha_i, \beta_i) + k_\tau \times \tau^0(\alpha_i, \beta_i) \right. \\ &\quad \left. + k_d \times \left((x_i - x_i^{LV})^2 + (y_i - y_i^{LV})^2 \right) \right) \end{aligned} \quad (23)$$

where $P_{ris}(\alpha_i, \beta_i)$ denotes the value of risk field, which can be calculated in (12). (x_i and y_i are the predicted longitudinal and lateral positions of vehicle i . LV refers to the near front vehicle of vehicle i . x_i^{LV} and y_i^{LV} denote the position of LV . $\tau^0(\alpha_i, \beta_i)$ means the TTC value between vehicle i and LV . k_τ and k_d are constant values.

The efficiency profit $J_e(\alpha_i, \beta_i)$ is derived as follows

$$J_e(\alpha_i, \beta_i) = v_i^{x,ave} \quad (24)$$

where $v_i^{x,ave}$ represents the average longitudinal speed in prediction time of vehicle i .

The platoon integration profit $J_{it}(\alpha_i, \beta_i)$ refers to the performance of maintaining the integrity of the platoon formation during the platoon reorganization process, which can be represented by traffic entropy, shown in (25).

$$J_{it}(\alpha_i, \beta_i) = -n_S \sum_{j=1}^l \left(\frac{p_j}{n_S} \ln \left(\frac{p_j}{n_S} \right) \right) \quad (25)$$

where n_S is the number of vehicles in coalition S , l denotes the number of lanes in roads, p_j is the number of the j -th lane, including both ego vehicles and background vehicles.

The tracking profit J_{tr} is characterized by the position and speed error of vehicles within ego platoon, denoted as follows

$$J_{tr}(\alpha_i, \beta_i) = \sum_{j=1}^{n_S-1} ((x_j - x_{j+1} - d_{target}) + k_y(y_j - y_{j+1}) + k_v(v_j - v_{j+1})) \quad (26)$$

where x_j , y_j , v_j are the lateral position, longitudinal position, and speed of the j -th vehicle in the platoon, x_{j+1} , y_{j+1} , v_{j+1} are the lateral position, longitudinal position, and speed of the $(j+1)$ -th vehicle in the platoon. k_y and k_v are weights for longitudinal position and speed control errors.

C. Platoon Disposition Index (PDI)

To further improve the performance of the platoon reorganization process, decrease the reorganization time, and enhance reorganization efficiency, we propose the Platoon Disposition Index (PDI) based on graph theory.

The PDI provides a comprehensive assessment of the traffic environment by representing both the spatial distribution of platoon vehicles and the difficulty of transitioning to the ideal formation. Unlike conventional error metrics that indirectly reflect platoon distribution through longitudinal or lateral errors and neglect surrounding traffic, the PDI explicitly incorporates both intra-platoon positions and the distribution of nearby vehicles. It consists of a temporal cost, defined as the discrete steps needed to reach the target, and a spatial cost, representing the geometric distance vehicles must traverse. By aggregating these components, the PDI captures both temporal and spatial challenges of redistribution and can be directly embedded into reward or objective functions to accelerate and regularize the reconfiguration process.

The PDI is calculated based on graph theory. Thus, we simplifies the road into a node map, where each vehicle or obstacle occupies a node. The schematic diagram of road node division is shown in Figure 6. d_n is the distance between

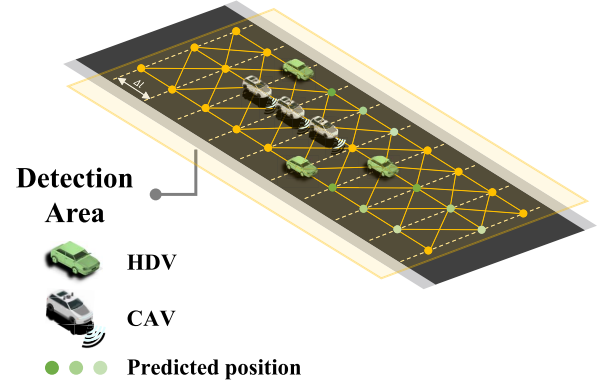


Fig. 6. Diagram of road node division.

nodes, $d_n \in [d_n^{\min}, d_n^{\max}]$. The values of $d_n^{\min}$ and $d_n^{\max}$ are related to the vehicle's dimensions. In this article, $d_n^{\min} = 10$ m and $d_n^{\max} = 20$ m.

Before calculating PDI, The following assumptions are made for the nodes:

- i) The node occupied by the first vehicle of the platoon is considered the starting point.
- ii) The node occupied by the last vehicle of the platoon is considered the end point.
- iii) Nodes occupied by background vehicles or obstacles are considered infeasible nodes.
- iv) Only adjacent nodes can be connected to form an edge. the adjacent nodes are defined in Definition 4.
- v) The weight of an edge is defined by both the distance between the two nodes and the lanes in which the nodes are located. Its physical meaning is the difficulty for a vehicle to move from one node to another, named Equivalence Distance (ED), is defined in Definition 5.

Definition 4 (Adjacent Nodes): Node $n_i \in N_n$ and node $n_j \in N_n$. N means the set of all nodes of the map. L_i and L_j are the lanes that n_i and n_j belong to. x_i and x_j are the longitudinal positions of n_i and n_j . If a vehicle occupies a node, the node is located at the center of the vehicle's rear axle. If a node is occupied by an obstacle, the node is located at the center of the physical outline of the obstacle. If $L_i = L_j$ and $|x_i - x_j| < d_n^{\max}$, then n_i and n_j are adjacent nodes. If $|L_i - L_j| = 1$ and there exists a n_k such that $|x_i - x_j| \leq |x_k - x_i|$, then n_i and n_j are adjacent nodes.

Definition 5 (Equivalence Distance): E_{ij} denotes the edge between node n_i and node n_j . The ED_{ij} is the weight of E_{ij} . The value of w_{ij} is given by (27). The edge distance (ED) combines the relative spatial distance $\frac{\tilde{d}_{ij}}{D}$ and the lane-changing difficulty $|L_i - L_j|$, where a higher minimum number of lane changes implies greater delay under dense traffic, thus indicating higher traversal difficulty [41].

$$E_{ij} = \frac{\tilde{d}_{ij}}{D} + k_l |L_i - L_j| \quad (27)$$

where $\tilde{d}_{ij}$ refers to the Euclidean distance between n_i and n_j , k_l is a constant value. In this article, $k_l = 10$.

Based on the above assumptions i) ~ v), the goal is to find the shortest path in terms of equivalent distance from the

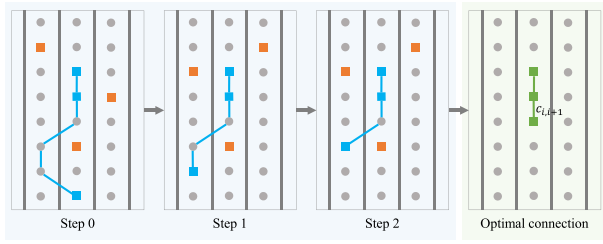


Fig. 7. PDI calculation process.

starting point to the endpoint, with the constraint that the path cannot pass through nodes occupied by background vehicles or obstacles. The sum of the ED between all the nodes along this path is defined as the equivalent length of the path. We will use the equivalent length of the path as the PDI. The schematic of the PDI calculation process is shown in Figure 7.

The process of finding the shortest path is modeled as a MIP to solve for the PDI. All edges in the graph are set as 0-1 integer variables, denoted by u . The number of variables is n_u . The objective function is given by (28). The constraints are shown in (29).

$$J_{path} = \sum_{i=1}^{n_u} ED_i u_i \quad (28)$$

$$\begin{cases} \sum_{j=1}^{n_u} u_{ij} = 1, & \text{if } n_i = n_0 \\ \sum_{j=1}^{n_u} u_{ij} = 0, & \text{if } n_j = n_0 \\ \sum_{j=1}^{n_u} u_{ij} = 1, & \text{if } n_i = n_e \\ \sum_{j=1}^{n_u} u_{ij} = 0, & \text{if } n_j = n_e \\ \sum_{i=1}^{n_u} u_{ij} = \sum_{i=1}^{n_u} u_{ji}, & \text{if } n_j \neq n_0 \text{ and } n_j \neq n_e \end{cases} \quad (29)$$

The PDI can be computed using common MIP solving methods, such as branch and bound, etc. The advantages of PDI mainly lie in: When vehicles in the platoon are in the same position but the positions of surrounding vehicles differ, traditional error metrics cannot distinguish such situations. For example, when the platoon is cut in by other vehicles, the traditional error indexes will make conservative decisions and guide the platoon to drive behind the cut-in vehicle, maintaining its original lane, in order to minimize tracking error indexes. However, by adopting PDI, the optimal decision is to change lanes to the left, allowing the platoon to transform into a tighter configuration more quickly, thereby improving platoon efficiency and fuel economy.

VI. SIMULATION AND DISCUSSION

A. Simulation Settings

The simulation experiments in this study were conducted on a hardware platform comprising Ubuntu 22.04 operating system, an NVIDIA RTX 4080 GPU, an Intel Core i7-13700 CPU. The simulation platform is built on the basis of PyTorch framework and OpenAI Gym environment. In the simulator, actions determined by specific policies are translated into low-level steering and acceleration signals through a closed-loop PID controller. The vehicles in the simulator are built on the basis of kinematic models. The longitude and lateral decisions

TABLE I
HYPERPARAMETERS OF DPL MODEL

Symbol	Definition	Value
N_t	Total Training Steps	10^6
λ	Learning Rate	3^{-4}
γ	Discount Factor	0.85
B	Batch Size	64
ϵ	Clip Parameter	0.2
S_u	Number of Forward Steps	128
τ	Target Update Rate	0.0
c_v	Value Loss Coefficient	0.5
c_e	Entropy Term Coefficient	0.0

of HDVs are controlled by Intelligent Driver Model (IDM) and MOBIL models, respectively.

In our simulation, the vehicles in the background traffic flow are generated randomly, with different driving styles to simulate the uncertainty of the environment. Due to communication distance and latency constraints, the number of vehicles in a platoon is typically limited to 2 to 5. In our case, the number of platoon vehicles is 3. All vehicles in the platoon are equipped with advanced autonomous driving capabilities and V2X or DSRC communication technologies.

In the proposed GRDF model, we adopt the Multi-layer Perception (MLP) as the encoder and decoder, which contains one hidden layer and two fully connected (FC) layers. The size of hidden and FC layers is 256. The detailed hyperparameter is shown in Table I.

B. Case Studies

When driving in highway roads, the platoon faces risks mainly from two directions, longitudinal and lateral. The U.S. PATH truck-platooning project (2017) [42] and the European ENSEMBLE project (2022) [43] report that, in mixed-traffic environments, emergency collision-avoidance events and unexpected cut-ins by human-driven vehicles constitute the two highest-risk scenarios for truck platoons. The longitudinal risk means potential collisions coming from the same lane as platoon. For instance, the front vehicle suddenly decelerates and causes a rear-end collision, which occurs frequently in high-speed traffic. The lateral risk indicates possible collisions caused by lateral adjacent vehicles as some drivers may take aggressive lane-change or overtake behaviors in order to pursue higher speed, which may lead to corner collision unluckily. Especially in ramp merging area, continuously vehicles merge into the merging lane of the main road, which brings about temporary congestion in merging lane. When platoon drives in uninterrupted near lane, the vehicles in merging lane is likely to cut into the platoon, breaking platoon formation and increasing lateral collision risk.

The emergency hard-braking avoidance and human-driver cut-in scenarios were selected as canonical test cases and generalized into concrete scenario instances for simulation. The key scenario variables comprise platoon cruising speed, leading-vehicle speed, cut-in vehicle speed, minimum gap between platoon vehicles and the leading vehicle, and minimum gap between platoon vehicles and the cut-in vehicle. Their respective sampling ranges are 20–25 m/s, 12–17 m/s,

12–17 m/s, 5–15 m, and 5–15 m. To increase scenario severity, at the instant a lateral or longitudinal hazardous event occurs the TTC between platoon vehicles and the obstacle vehicle is constrained to the interval 0.5–2.5 s.

Hence, in this article, we organize two simulation cases of longitudinal and lateral driving risks to evaluate the performance of proposed model, respectively.

1) *Case 1: Lateral Driving Risk*: In Case 1, we simulated a typical lateral high-risk scenario in a three-lane highway environment with ramp merging area. Lateral high risk refers to situations where the platoon's original formation is disrupted by lateral interference from other vehicles or objects. At highway ramp merging points, unclear right-of-way between main road and ramp vehicles significantly raises the driving risk. Additionally, the continuous flow of ramp vehicles into the rightmost lane often causes congestion in that lane. To improve driving efficiency, vehicles in the rightmost lane are more likely to change lanes into the middle lane. If the controlled vehicle platoon happens to be traveling in the middle lane, the platoon's substantial spatial occupation makes it highly likely for vehicles from the rightmost lane to cut into the platoon. This increases the risk of collision within the platoon significantly.

At first, the platoon drives in tight line formation in the middle lane to decline air resistance. The distance between the rear axle center of the lead vehicle and the rear axle center of the following vehicle is 10 m. The length of vehicle body is 5 m.

To validate the advantages of the proposed multi-lane platoon reorganization strategy, we compared it with two cooperative lane-changing methods for platooning. The SiPLC proposed in 2018 [44] and SuPLC proposed in 2022 [11], are selected as baseline models. The SiPLC allows platoon to change lanes holistically, while maintaining original formation at the same time. This will lead to conservative lane change behaviors. The platoon can only change lane as the premise of sufficient longitudinal spacing, which is hard to be satisfied in real highway environment, especially under emergency conditions. Besides, the SuPLC permits platoon to take successive lane-change, cutting through the traffic like a snake [11]. This method occupies smaller road space but extends the lane changing time window, which is difficult to implement in emergency environments.

To visually demonstrate the differences in performance across various control models, we used scenario slice diagrams to illustrate the platoon's driving behavior. Relevant video materials can be found on the website [githubhttps://github.com/KongAAAAAJ/GRDF](https://github.com/KongAAAAAJ/GRDF). The platoon's performance in the same scenario after applying the GRDF, SiPLC, and SuPLC methods is shown in Figure 8 and 9.

Figure 8 illustrates scenario slice diagrams of platoon driving under the baseline methods. The left side of Figure 8 shows platoon's driving behavior under SiPLC. At $t=1.2$ min, the platoon initiated a left lane-changing maneuver, with all vehicles in the platoon changing lanes simultaneously. As the number of merging ramp vehicles in the rightmost lane increased, a human-driven vehicle in the rear of the rightmost

TABLE II
PERFORMANCE EVALUATION OF CASE 1

Index	SiPLC	SuPLC	GRDF
Collision Rate	0.884	0.965	0.021
Average Speed (m/s)	24.47	24.08	25.42
Minimum TTC (s)	3.21	3.39	7.23
Average Distance (m)	7.39	9.57	9.09

lane exhibited an intention to change lanes to the left. At $t=7.5$ min, this vehicle cut into the space between the first and second vehicles of the autonomous platoon. The second vehicle of the platoon performed an emergency brake to avoid a collision. However, the third vehicle failed to decelerate in time, resulting in a rear-end collision with the second vehicle.

The right side of Figure 8 shows platoon's driving behavior under SuPLC. At $t=2.7$ min, the platoon initiated a left lane-changing maneuver, while the last vehicle in the rightmost lane also exhibited an intention to change lanes to the left. Since the platoon adopts a serpentine lane-changing method, the lane-changing times for the platoon's vehicles gradually increase from the first vehicle to the last. At $t=7.2$ min, a human-driven vehicle cut into the space between the first and second vehicles of the platoon. By this time, the first vehicle had already completed the lane change, while the second vehicle was still in the process of changing lanes. Due to the human-driven vehicle's cut-in, the second vehicle failed to avoid the collision in time.

Figure 9 shows the vehicle platoon's driving behaviors after applying the proposed GRDF method. As seen in Figure 9, at $t=3$ min, a human-driven vehicle cuts into the platoon. At $t=5$ min, the platoon detects the cut-in behavior of another vehicle, and the second vehicle, most affected by the cut-in, initiates a left lane change to avoid a collision. After the second vehicle changes lanes, between $t=10$ min and $t=16$ min, the remaining vehicles in the platoon sequentially change lanes with the aim of restoring the original platoon formation, ultimately successfully completing the platoon reorganization process.

This demonstrates that the SiPLC and SuPLC methods cannot effectively handle the vehicle cut-in risk during cooperative lane-change process of the platoon. These methods struggle to address unexpected dangerous scenarios, exhibiting poor robustness and insufficient safety. However, the GRDF method can autonomously organize the vehicles to complete the platoon reorganization process in an orderly manner when facing the risk of cut-in by other vehicles, effectively preventing collisions and enhancing the safety and robustness of platoon driving.

To evaluate the performance of the proposed GRDF method in terms of driving efficiency and safety, we conducted 100 simulation experiments under random traffic flow, focusing on the lateral risk scenarios. The evaluation metrics selected include collision rate, average platoon speed, average inter-vehicle distance within the platoon, and the minimum TTC of the platoon vehicles. The results of these metrics are shown in Table II.

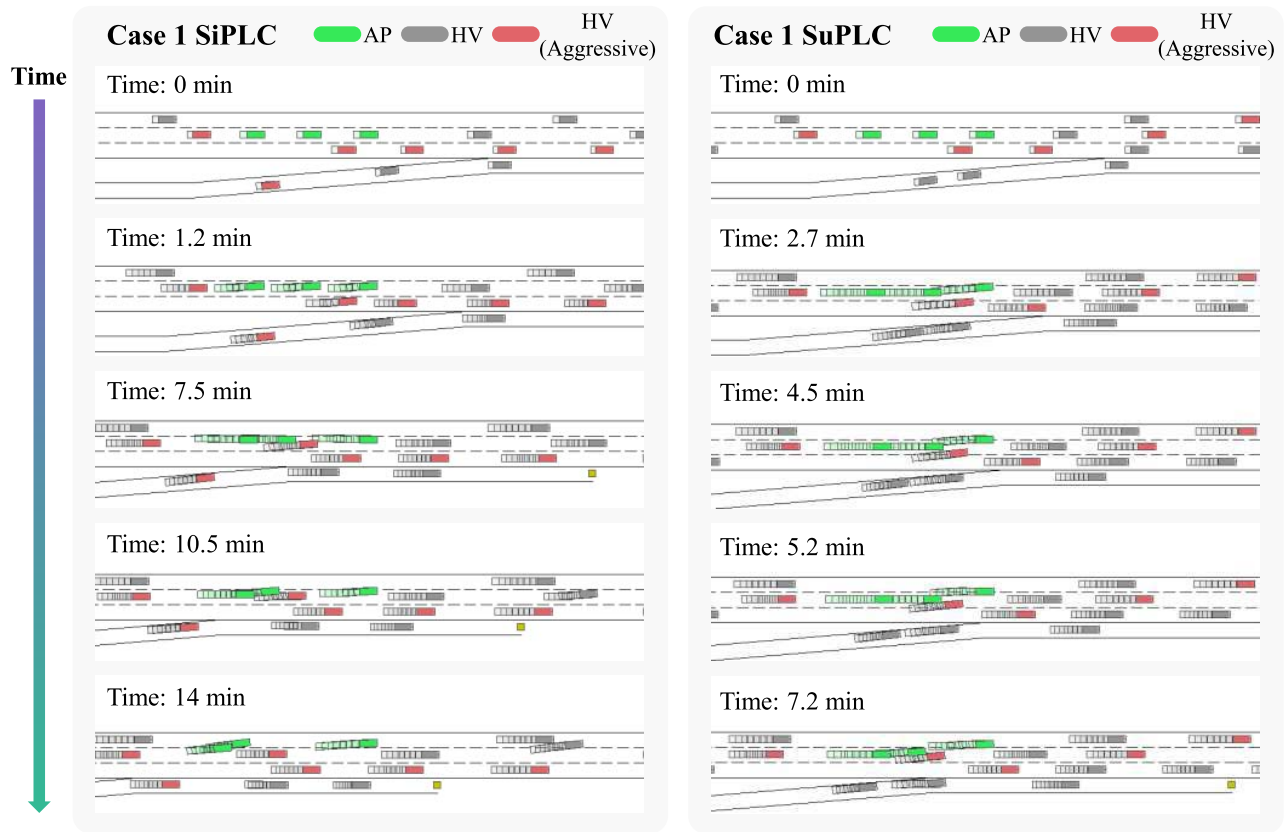


Fig. 8. The scenario slice diagram of SiPLC and SuPLC.

As shown in Table II, after applying the SiPLC and SuPLC methods, the platoon experiences a relatively high collision rate in the cut-in scenarios, with values of 0.884 and 0.965, respectively. However, after applying the GRDF method, the platoon's collision rate drops to 0.021, representing a reduction of 97.6% and 97.8%, significantly enhancing collision safety. From the Minimum TTC metric, after applying the baseline models, the Minimum TTC values are 3.21 s and 3.39 s, respectively, while the TTC value for GRDF is 7.23 s, which is 2.25 times and 2.13 times higher than the baseline models. The improvement in Minimum TTC indirectly reflects the enhanced safety of the platoon's driving. In addition, the average speed of the platoon increased from 24.47 m/s and 24.08 m/s to 25.42 m/s, representing an improvement of 3.9% and 5.6%, respectively.

The data distribution of metrics are illustrated in box plots, as shown in Figure 10 and Figure 11. The Average Speed and Minimum TTC values achieved by the GRDF model both surpass those of the baseline models. GRDF not only enhanced the safety of the platoon but also slightly improved its driving efficiency. This indicates that after applying GRDF, the platoon made better decisions, significantly improving safety without sacrificing efficiency, and effectively addressing lateral high-risk scenarios such as vehicle cut-ins.

The Average Distance metric refers to the average following distance between vehicles within the platoon, with an ideal following distance of 10 meters. SiPLC causes the vehicles to change lanes simultaneously, resulting in the shortest

Average Distance of 7.39 m. SuPLC, which has the vehicles change lanes in a serpentine manner, leads to the longest Average Distance of 9.57 m. During the platoon reorganization process, the following distance inevitably increases. However, after applying the GRDF strategy, the Average Distance is 9.09 m, which is still close to the ideal following distance. This indicates that, in most cases, the platoon maintains an ideal following distance and formation, implementing the reorganization strategy only when the risk level is higher.

2) *Case 2: Longitudinal Driving Risk*: In Case 2, we simulated a typical longitudinal risk scenario for platooning in a three-lane highway environment. The longitudinal risk in this case is brought about by the rapid deceleration behavior from the vehicle in front of the whole platoon. When the front vehicle makes a sharp deceleration, the longitudinal collision risk for rear platoon will be greatly increased. The platoon needs to avoid the obstacle vehicle and reduce the loss of traffic efficiency as much as possible.

To validate the superiority of the proposed graph-theory-and-game-based formation reorganization method (GRDF-GT), we compared GRDF-GT, GRDF, and a reinforcement-learning-and-rule-based formation reorganization method (RRL), closely observing changes in relevant performance metrics.

To eliminate uncertainties, the simulation experiments were repeated 100 times under random traffic flow conditions. The results were then subjected to statistical analysis, and the key performance metrics are summarized in Table III.

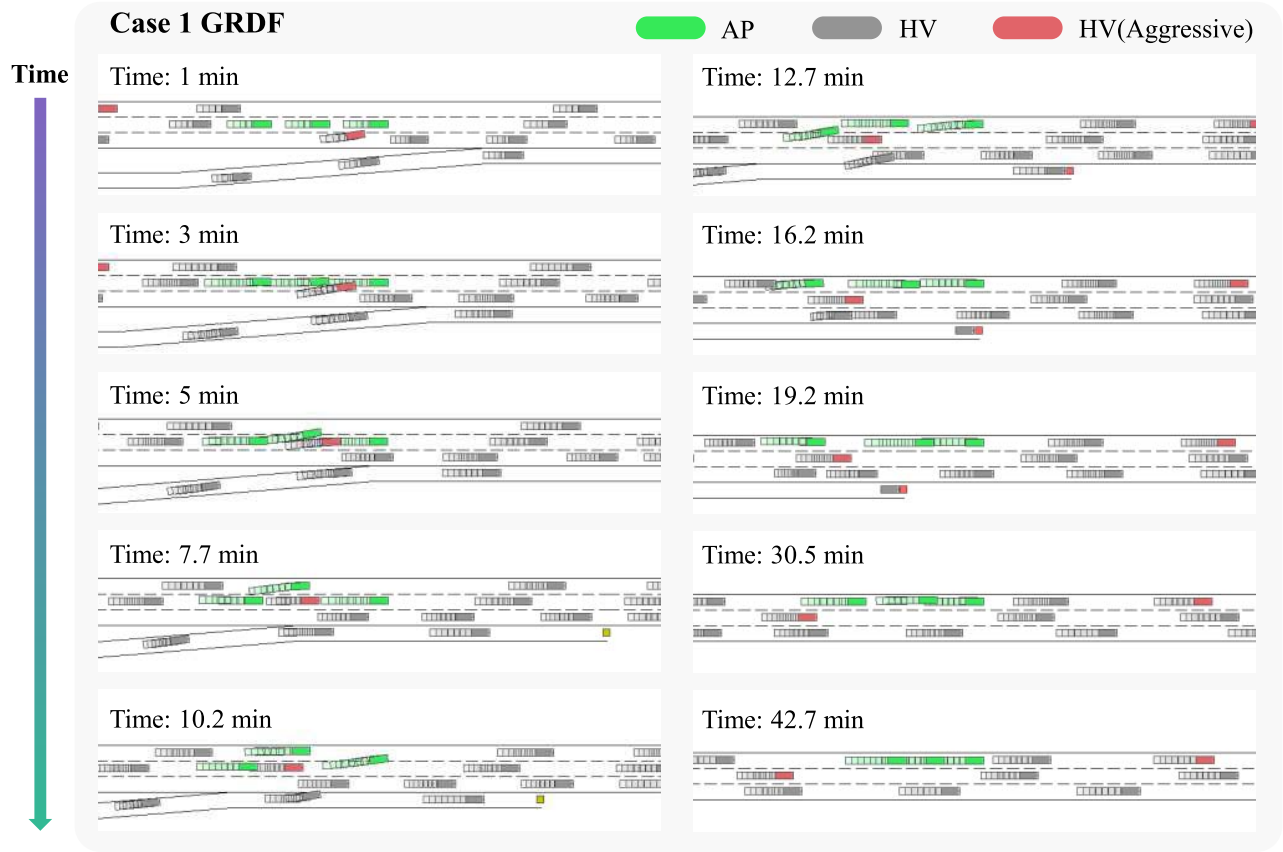


Fig. 9. The scenario slice diagram of GRDF.

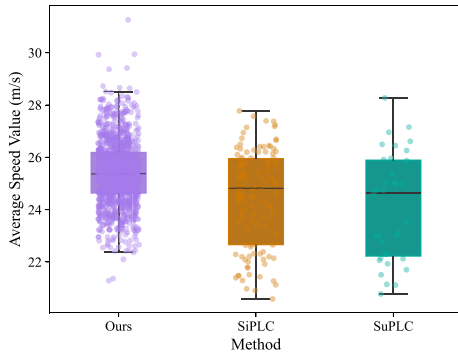


Fig. 10. The box plot of average speed.

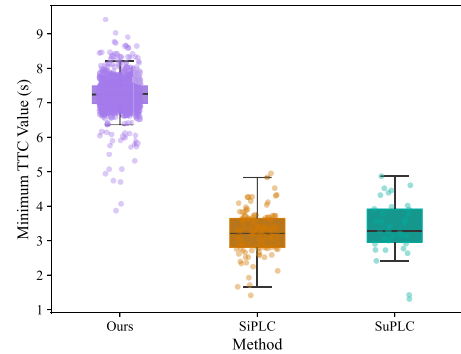


Fig. 11. The box plot of minimum TTC.

TABLE III
PERFORMANCE EVALUATION OF CASE 2

Index	RRL	GRDF	GRDF-GT
Collision Rate	0.21	0.01	0
Average Speed (m/s)	24.27	26.08	24.91
Minimum TTC (s)	6.50	4.13	4.08
Average Distance (m)	12.58	10.19	9.74
Formation Success Rate	0.80	0.88	0.94
Formation Time (min)	6.45	6.16	4.56

From Table III, it can be observed that both GRDF and GRDF-GT significantly reduce the collision rate, from 0.21 to 0.01 and 0. Although the Minimum TTC indicator slightly

decreases from 6.50 s to 4.13 s and 4.08 s, it remains within a safe range. Overall, the application of GRDF and GRDF-GT substantially enhances the safety of the platoon's driving performance. The Average Speed indicator also increases from 24.27 m/s to 26.08 m/s and 24.91 m/s, respectively, indicating an improvement in platoon driving efficiency while ensuring safety.

The indicators Average Distance, Formation Success Rate, and Formation Time reflect the capabilities of the platoon during the reorganization process.

i) Formation Success Rate refers to whether the platoon successfully restores its original formation and completes the reorganization process within 15 min. A higher value for this

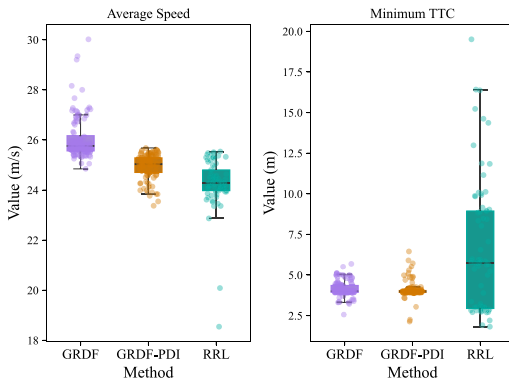


Fig. 12. The box plot of average speed and minimum TTC.

indicator signifies stronger capability in executing formation reorganization.

ii) Formation Time measures the time required for the platoon to restore its original formation from the initiation of the reorganization process. A lower value for this indicator reflects higher efficiency in the platoon's reorganization process.

As shown in Table III, the Average Distance decreased from 12.58 m and 10.09 m to 9.74 m with the application of GRDF-GT. This indicates that during the reorganization process, the platoon maintained a smaller inter-vehicle distance, enhancing the overall driving efficiency. Furthermore, the Formation Success Rate was highest with GRDF-GT at 0.94, followed by GRDF at 0.88, and RRL at 0.80. In addition, the Formation Time was minimized with GRDF-GT, achieving 4.56 min, significantly lower than 6.45 min for RRL and 6.16 min for GRDF. These results demonstrate that GRDF-GT effectively enhances the platoon's formation transformation capabilities and reorganization efficiency, improving its adaptability and robustness in risk-prone scenarios.

Meanwhile, the distribution of the Average Speed and Minimum TTC indicators is presented in the form of box plots in Figure 12. It can be seen that after applying the GRDF method, the data points for the Average Speed indicator of the vehicle platoon are mostly located above those of the RRL method, demonstrating a significant advantage in terms of efficiency. In contrast, the distribution of the Minimum TTC indicator for the RRL method is quite wide, indicating that the decision-making safety of RRL is unstable across different traffic flow scenarios. There are even many data points below 2.5 s, further decreasing safety. However, after applying the GRDF and GRDF-GT models, the distribution of the platoon's Minimum TTC indicator is more concentrated, with the vast majority of data points above 3.0 s, indicating a higher level of decision-making safety.

To provide a detailed presentation of the distribution density of various indicators during the platoon reorganization process, violin plots were used to analyze the Average Distance and Formation Time indicators. From Figure 13, it can be observed that the Average Distance indicator for the RRL model is approximately uniformly distributed between 7 and 18 m. This suggests that the model is quite sensitive to changes in scene parameters, and the platoon maintains a relatively tight formation only in certain simpler scenarios. In

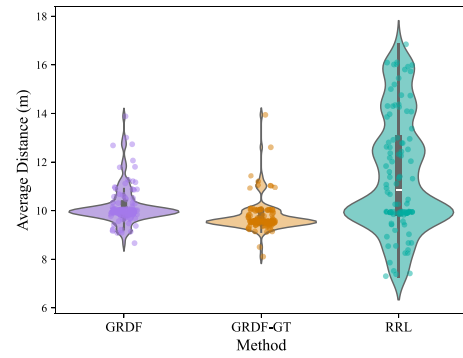


Fig. 13. The violin plot of average distance.

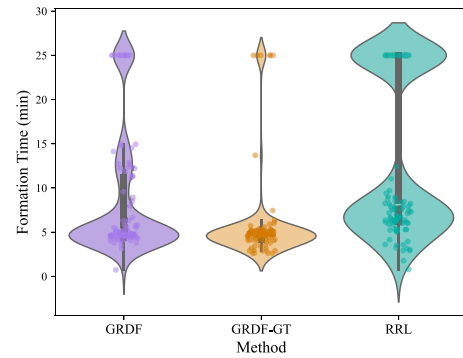


Fig. 14. The violin plot of formation time.

contrast, both the GRDF and GRDF-GT models exhibit a more concentrated distribution of the Average Distance indicator, indicating that their decision-making methods have stronger adaptability to varying traffic scenarios. From Figure 14, it can be observed that the Formation Time data for the RRL model has a wide distribution range, strongly correlating with changes in scene parameters. A significant portion of the data is distributed at the upper part of the graph, around $t = 25$ min, indicating that in many scenarios, the platoon does not complete the reorganization process within the specified time. In contrast, the Formation Time data for the GRDF-GT model is more concentrated and shows lower values, highlighting that the GRDF-GT model significantly improves platoon reorganization efficiency and enhances its adaptability to varying environments.

C. Discussion

For real-vehicle implementation, the proposed algorithm is deployed in autonomous platoon systems equipped with advanced sensors, including cameras, millimeter-wave radar, and LiDAR, enabling obstacle detection and estimation of object position, shape, and velocity. The algorithm runs on the lead vehicle, and its outputs are transmitted in real time to other platoon members via LTE-V2X. A bidirectional communication channel exists between any two vehicles, with the active communication pattern determined by vehicle roles. Specifically, leaders exchange information bidirectionally; a follower transmits its local state to the leader within the same sub-platoon; and when both vehicles are followers in the same

sub-platoon, the rear vehicle receives the front vehicle's state information.

Simulation results show that the synchronization requirements for multi-vehicle cooperative control are satisfied. By employing a dynamic pruning strategy, the system response period is maintained at or below 0.1 s, meeting low-frequency real-time demands, although further reduction is desirable for high-frequency scenarios. In typical vehicular networks, one-way latency is below 10 ms and available bandwidths (≥ 10 Mbps) suffice for decision-process data throughput [45], while measured packet loss rates of 3–8% indicate a need for improved communication reliability. From a computational perspective, modern in-vehicle platforms (e.g., NVIDIA Drive AGX Orin with ≥ 200 TOPS) provide sufficient processing power for real-time algorithm execution.

Overall, computational capacity and nominal bandwidth are adequate, and low-frequency real-time performance meets targets, but communication robustness and algorithmic resilience to disturbances, such as GPS outages, require enhancement to ensure dependable operation in complex real-world conditions.

VII. CONCLUSION

This article proposes a decision-making and control framework for multi-lane platoon formation reorganization, aiming to enhance the platoon's ability to respond to dynamic emergency environment. The framework divides the reorganization process into the upper platoon-level distribution layer and the lower vehicle-level collaborative decision-making layer. In the upper layer, a reinforcement learning-based platoon formation decision algorithm is introduced, which uses a risk potential field to establish explicit risk evaluation metrics. A reward function tailored to the platoon reorganization process is designed to guide exploratory actions. In the lower layer, a coalition game algorithm is used to model cooperative relationships among vehicles. An innovative PDI is proposed to accelerate the platoon reorganization process and improve platoon driving efficiency. In addition, the lateral and longitudinal high-risk scenarios are designed based on the directional distribution of risk sources to test the proposed method. After the testings in random traffic flows, the results show that compared to the baseline model, the proposed method reduces the collision rate in longitudinal risk scenarios from 0.884 and 0.965 to 0.021, increasing the average speed from 24.08 m/s and 24.47 m/s to 25.42 m/s. Both platoon driving safety and efficiency are improved. Besides, the proposed PDI reduces the platoon reorganization time from 6.45 min and 6.16 min to 4.56 min, significantly enhancing the efficiency of platoon reformation and configuration transitions.

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